

Surgery for soft tissue sarcoma

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About soft tissue sarcoma surgery

[Soft tissue sarcomas](#) are rare, so surgery is done by a surgeon who specialises in treating them. The surgeon is part of a [multidisciplinary team \(MDT\)](#) and based in a specialist sarcoma unit.

You may have a lump removed and only find out it is a soft tissue sarcoma after the surgery. In this situation, you will be referred to a sarcoma specialist who is part of an MDT after your initial surgery.

The aim of most sarcoma surgery is to remove all of the tumour. Usually, an operation called a wide local excision is done. This means the surgeon removes the tumour, along with an area (margin) of healthy, cancer-free tissue around it.

After the operation, the pathologist examines the tissue from the margin. If there are cancerous cells in the margin, you may need another operation to remove more tissue. Making sure the margins are clear reduces the risk of the cancer coming back.

Types of surgery for soft tissue sarcoma

It is difficult to give general information about sarcoma surgery, because the type of operation you have will depend on where in your body the sarcoma is. In this information, we explain the types of surgery used for people with a soft tissue sarcoma in different parts of the body.

Some types of surgery for sarcoma are minor and only involve removing a small amount of tissue. Some operations are major and may affect how that area of the body works, or how it looks.

Before surgery, you may be referred to a rehabilitation team. They will help you recover from the operation and keep as much function as possible. If the surgery is very minor, you may not need the help of the rehabilitation team.

Your surgeon and specialist nurse will discuss your operation and rehabilitation with you.

Wide local excision

The aim of most sarcoma surgery is to remove all of the tumour. Usually, an operation called a wide local excision is done. This is when the surgeon removes the tumour, along with an area of healthy, cancer-free tissue around it. This area is called a margin.

After the operation, a specialist called a pathologist examines the tissue from the margin. If there are cancerous cells in the margin, you may need another operation to remove more tissue. Making sure the margins are clear reduces the risk of the cancer coming back.

The type of operation you have will depend on where the sarcoma is. You may have other treatments before or after [surgery](#) or both.

Surgery to the chest or tummy (abdomen)

[Surgery](#) is usually the main treatment for soft tissue sarcomas in the chest and tummy (abdomen). This area of the body is called the trunk. The type of operation will depend on the position of the tumour. Your surgeon will discuss this with you before the operation.

If the tumour is in the tummy, it can be difficult to remove it along with an area of healthy tissue (margin). This is because the tumour may be very close to important organs, like the kidneys and liver, or main blood vessels.

You may have [radiotherapy](#) after an operation to remove a sarcoma in the chest or tummy. This is to try to make sure any remaining cancer cells are destroyed.

Surgery to the arms or legs (limbs)

It is usually possible to remove a soft tissue sarcoma in an arm or leg without having to remove the whole limb. This is called limb-sparing surgery. It is often done by using a combination of surgery, [radiotherapy](#) and occasionally [chemotherapy](#).

We have more detailed information about what to expect before and after [limb-sparing surgery](#).

Rarely, the best treatment option is to remove the affected limb to make sure all the cancer has been removed. This is called [amputation](#).

Surgery to the head and neck

The type of surgery you have will depend on the exact position and size of the tumour. Your surgeon will discuss this with you before the operation.

It is important that you understand what is going to be removed and how this will affect you after the operation.

Changes to speech or swallowing

Your surgeon will do everything possible to reduce any effects surgery may have on the way you:

- speak
- swallow
- look.

Sometimes there is a risk your speech, swallowing or eating could be affected for a time after surgery. In this situation you will see a:

- speech and language therapist (SLT), who will teach you exercises to help you communicate
- dietitian, who will give you advice about eating and make sure you get enough nutrition.
- They help to prepare you for surgery and continue to support you after surgery.

If the operation involves removing tissue used for speech and swallowing, or affects your appearance, you may have reconstructive surgery at the same time.

A plastic surgeon takes tissue from another part of the body, such as the forearm, thigh or chest. This is used to replace the tissue they take from the head and neck.

Reconstructive surgery

Depending on the size and position of the sarcoma, the surgeon may have to remove a large area of tissue. This may include important nerves and blood vessels. You may need to have surgery to reconstruct the area. This is done at the same time as your operation, usually by a plastic surgeon.

The aims of reconstructive surgery are to:

- make the area look and work as naturally as possible
- pad and protect important tissue nearby
- help the wound to heal.

The operation will depend on where in your body the sarcoma is and how much tissue the surgeon needs to remove. If it is a smaller area, the surgeon may be able to join the 2 edges together again. For larger areas, they may need to use a skin graft or tissue flap.

Skin grafts

A [skin graft](#) is a very thin layer of skin. The surgeon takes a layer of skin from another part of the body called the donor site. They put this over the area where the cancer was.

Skin is often taken from the outer thigh. If a skin graft is needed for the face, surgeons usually take skin from behind the ear or the neck. This is done to try to get a good skin colour match.

After the operation, you normally have a dressing over the area. This presses the graft down to help create a good blood supply from the blood vessels underneath.

Tissue flaps

Sometimes, plastic surgeons need to use a specialist technique to repair the operation site. They use a slightly thicker layer of skin and tissue than for a skin graft. This is called a [tissue flap](#). Tissue flaps include fat and muscle. The different types of tissue flap include the following:

- A local flap uses tissue from an area close to where the cancer has been removed.
- A pedicle flap uses tissue from an area further away, but close enough to keep the flap connected to its original blood supply.
- A free flap uses tissue that has been completely removed from another part of the body, along with its blood vessels.

The surgeon and specialist nurse will explain your operation and answer any questions you have.

Surgery to remove sarcoma that has spread

Surgery is sometimes used to remove a sarcoma that has spread to other parts of the body or come back after treatment. An operation is usually only possible if:

- the tumours are small
- there are only a few tumours.

It is mainly done when the sarcoma has [spread to the lungs](#). But it might be used for tumours in other parts of the body. If you need to have this operation, your doctor will talk to you about it.

About our information

References



Below is a sample of the sources used in our soft tissue sarcoma information. If you would like more information about the sources we use, please contact us at cancerinformationteam@macmillan.org.uk

Gronchi A, Miah AB et al. Soft tissue and visceral sarcomas: ESMO-EURACAN-GENTURIS Clinical practice guidelines for diagnosis, treatment and follow-up. *Annals of Oncology*, 2021; 32, 11, 1348-1365 [accessed May 2022].

Casali PG, Blay JY et al. Gastrointestinal stromal tumours: ESMO-EURACAN-GENTURIS Clinical practice guidelines for diagnosis, treatment and follow-up. *Annals of Oncology*, 2022; 33,1, 20-33 [accessed May 2022].

Reviewers



This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer. It has been approved by senior medical editor [Fiona Cowie](#), Consultant Clinical Oncologist.

Our cancer information has been awarded the [PIF TICK](#). Created by the Patient Information Forum, this quality mark shows we meet PIF's 10 criteria for trustworthy health information.

The language we use

We want everyone affected by cancer to feel our information is written for them.

We want our information to be as clear as possible. To do this, we try to:

- use plain English
- explain medical words
- use short sentences
- use illustrations to explain text
-

structure the information clearly

- make sure important points are clear.

We use gender-inclusive language and talk to our readers as 'you' so that everyone feels included. Where clinically necessary we use the terms 'men' and 'women' or 'male' and 'female'. For example, we do so when talking about parts of the body or mentioning statistics or research about who is affected.

You can read more about how we produce our information [here](#).

Date reviewed

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ⓘ This content is currently being reviewed. New information will be coming soon.

Our cancer information meets the [PIF TICK](#) quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about [how we produce our information](#).

How we can help

Macmillan Support Line

The Macmillan Support Line is a free and confidential service for people living with and affected by cancer. If you need to talk, we'll listen.



Call us free on 0808 808 0000

8am to 8pm, 7 days a week. You can talk to our cancer specialists about anything.



Email the Macmillan Support Line

Email the Macmillan Support Team if you have any questions.

Chat online

Available 8am to 7.45pm, 7 days a week. For confidential, real-time support if you don't want to call.



Online Community

Chat online anonymously to others who understand what you are going through. Our community is available 24/7 and has dedicated forums where you can get advice and ask our experts.



Find cancer support services in your area

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.

Get help